

Write Inequalities

Lesson 7-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x > 5$ means x can be 6, 7, 8, ... — any number greater than 5

Inequality

A math sentence that compares two sides with $<$, $>$, $\leq$, or $\geq$.

$8 > 3$ — eight is larger than three

Greater than

The $>$ sign, showing one number is bigger.

$2 < 7$ — two is smaller than seven

Less than

The $<$ sign, showing one number is smaller.

'At least 10 players' $\rightarrow p \geq 10$ (could be 10, 11, 12, ...); 'At most 5 tries' $\rightarrow t \leq 5$ (could be 5, 4, 3, ...)

At least / At most

'At least' means $\geq$. 'At most' means $\leq$.



X

stands for a number

In $x > 5$, the letter x stands for any number greater than 5.

Variable

A letter that stands for an unknown number.



#

number

"No more than 8 people" means $\text{people} \leq 8$.

No more than

A phrase meaning less than or equal to ($\leq$).

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can write inequalities to represent real-world situations.

1 Notice

Detective Navarro is narrowing down suspects.

2 Key idea

I can write inequalities to represent real-world situations.

3 Words I'll use

Inequality

Greater than

Less than

At least / At most

Variable

No more than



Think About It

- What does 'at least 18' mean about the suspect's age?
- Could the suspect be exactly 18?
- How is an inequality different from an equation?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

I do — watch

Follow each step as your teacher solves it.

Problem: Which inequality represents: 'The temperature is less than 40 degrees'?

- A. $t < 40$
- B. $t > 40$
- C. $t \leq 40$
- D. $t = 40$

Step 1 'Less than' means $<$, so the inequality is $t < 40$.

Answer: A. $t < 40$

 We do — together

Solve this one with your class using the same steps.

Problem: Which inequality represents: 'You must be at least 48 inches tall to ride'?

A. $h \geq 48$

B. $h > 48$

C. $h < 48$

D. $h \leq 48$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Which inequality represents: 'A store can hold no more than 75 people'?

A. $p \leq 75$

B. $p < 75$

C. $p \geq 75$

D. $p > 75$

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

A witness says the suspect is 'at least 18 years old,' written as $a \geq 18$. Why does the detective use an inequality here instead of an equation like $a = 18$?

Level 1 support

Start here: The age could be 18 or any number greater, so it is not a single value.

 **Need a hint?**

Sentence starters (optional)

An inequality fits better than an equation because ____.

Una desigualdad encaja mejor que una ecuación porque ____.

'At least 18' means the age can be ____.

'Al menos 18' significa que la edad puede ser ____.

Word bank: inequality greater than less than at least / at most

Listen for: Students explain that 'at least 18' allows many values (18 and up), so an inequality with $\geq$ captures it while an equation would force one value.

Level 2

How would the meaning change if the clue used $a > 18$ instead of $a \geq 18$? Justify whether an 18-year-old still fits.

- With $a > 18$ an 18-year-old would ____ because ____.
- The line under the symbol means ____.

EXPLORE

How do the phrases 'at least' and 'at most' change which inequality symbol you choose compared to 'more than' and 'fewer than'?

Level 1 support

Start here: 'At least' and 'at most' include the number; 'more/fewer than' do not.



Need a hint?

Sentence starters (optional)

'At least' uses ____ because it ____ the number, but 'more than' uses ____.

'Al menos' usa ____ porque ____ el número, pero 'más que' usa ____.

My inequality is ____ because the phrase says ____.

Mi desigualdad es ____ porque la frase dice ____.

Word bank: inequality greater than less than at least / at most

Listen for: Students match 'at least' to $\geq$ and 'at most' to $\leq$ (include the value), and 'more than'/'fewer than' to $>$ and $<$ (exclude the value).

Level 2

Compare 'no more than 8 exits' with 'fewer than 8 exits.' Could 8 exits be allowed in each case? Justify.

- 8 exits is allowed with ____ but not with ____ because ____.
- The difference is whether the number is ____.

CONNECT

Where do you see inequalities in real life, such as age limits or speed limits?

Level 1 support

Start here: Many real rules describe a range of allowed values, not one value.



Need a hint?

Sentence starters (optional)

I see an inequality when _____, and I would write _____.

Veo una desigualdad cuando _____, y escribiría _____.

This is an inequality, not an equation, because _____.

Esto es una desigualdad, no una ecuación, porque _____.

Word bank: inequality greater than less than at least / at most

Listen for: Students give a real example (ride height, speed limit, ticket age) and write an inequality with the correct symbol for a range of values.

Level 2

Critique: 'A speed limit of 25 mph means you must drive exactly 25.' Write the correct inequality and explain the real rule.

- This is wrong because a speed limit means _____.
- The correct inequality is _____ because _____.

Write About the Math

The Writing Revolution

I can explain my inequality using the words inequality, greater than, less than, and at least / at most.

1. Kernel Sentence subject + verb

Model: Inequality is a math sentence that compares two sides with $<$, $>$, $\leq$, or $\geq$.

Desigualdad es una oración matemática que compara dos lados con $<$, $>$, $\leq$ o $\geq$.

Write a kernel sentence about inequality. Use a subject and a verb.

Escribe una oración base sobre desigualdad. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Inequality matters in math

Desigualdad importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Inequality matters in math because ____.

Desigualdad importa en matemáticas porque ____.

but
pero

Inequality matters in math, but ____.

Desigualdad importa en matemáticas, pero ____.

so
entonces

Inequality matters in math, so ____.

Desigualdad importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about inequality.
Di un hecho verdadero sobre inequality.

Inequality ____.

Question
Pregunta

Ask a question about inequality.
Haz una pregunta sobre inequality.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about inequality.
Muestra entusiasmo sobre inequality.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with inequality.
Dile a un compañero qué hacer con inequality.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

An inequality fits better than an equation because ____.

Una desigualdad encaja mejor que una ecuación porque ____.

'At least 18' means the age can be ____.

'Al menos 18' significa que la edad puede ser ____.

'At least' uses ____ **because it** ____ **the number, but 'more than' uses** ____.

'Al menos' usa ____ *porque* ____ *el número, pero 'más que' usa* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A bus holds at most 40 people. Which inequality shows the number p of people allowed?

A. $p \leq 40$

B. $p \geq 40$

C. $p > 40$

D. $p < 40$

Show your work:

2. Which inequality means 'a number n is less than 8'?

A. $n < 8$

B. $n \leq 8$

C. $n > 8$

D. $n \geq 8$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write two different real-world situations: one that uses $>$ and one that uses $\geq$.

Explain why the situations need different symbols even though both mean 'bigger.'

Sentence starter: Situation 1 ($>$): _____. This uses $>$ because _____. Situation 2 ($\geq$): _____. This uses $\geq$ because _____. The difference is that $\geq$ includes _____ while $>$ does not.

Show your work:

Reflect — Exit Ticket

Which inequality represents: 'A number n is at least 14'?

- A. $n \geq 14$
- B. $n > 14$
- C. $n \leq 14$
- D. $n < 14$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $h \geq 48$ — '*At least*' means *greater than or equal to*, so $h \geq 48$.
2. **You Try (notes):** A. $p \leq 75$ — '*No more than*' means *less than or equal to*, so $p \leq 75$.
3. **Try It 1:** A. $p \leq 40$ — '*At most 40*' means *40 or fewer*: $p \leq 40$.
4. **Try It 2:** A. $n < 8$ — '*Less than 8*' is $n < 8$.
5. **Exit Ticket:** A. $n \geq 14$ — '*At least 14*' means *14 or greater*, so $n \geq 14$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Inequality is a math sentence that compares two sides with $<$, $>$, $\leq$, or $\geq$.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).